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Venepuncture versus heel lance for blood sampling in term neonates (Review)

Martins G, Fiander M, Taddio A, Shah VS

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[Intervention Review]

Venepuncture versus heel lance for blood sampling in term neonates

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ABSTRACT

Rationale

Reducing procedural pain is a key priority in neonatal care, as even routine procedures for blood sampling can cause significant discomfort and pain to the neonate. This systematic review aims to compare venepuncture and heel lance for blood sampling, focusing on pain response, to guide clinical practice.

Objectives

To evaluate the benefits (degree of pain) and harms (need for more than one skin puncture) of venepuncture compared to heel lance in term neonates.

Search methods

We searched CENTRAL, MEDLINE, Embase, two trial registries and conference abstracts, and checked references of included studies and related systematic reviews up to 04 July 2025.

Eligibility criteria

We included randomized controlled trials (RCTs) and quasi-RCTs that compared pain response from venepuncture versus heel lance. Neonates could receive a sweet-tasting solution (such as glucose or sucrose), which is a commonly used analgesic in neonates for pain relief.

We excluded studies that reported on pain using physiological parameters in isolation, as they are reflective of overall clinical status and non-specific arousal.

Outcomes

Our outcomes of interest were pain reported using validated pain scores (measured using one or more of the following scales: Neonatal Facial Coding System, Neonatal Infant Pain scores, or the Premature Infant Pain Profile) and the need for more than one skin puncture.

Risk of bias

We used the Risk of bias 1 (RoB) tool to assess bias in the included RCTs.

Synthesis methods

We conducted a meta-analysis where possible by calculating standardized mean difference (SMD) and risk ratio (RR) with their 95% confidence intervals (CIs) for continuous and dichotomous outcomes, respectively. We used GRADE to assess the certainty of evidence for each outcome.

Included studies

We included eight studies with 826 neonates. The studies were performed in six high-, upper-middle, or low-income countries.

Synthesis of results

We judged the overall risk of bias in the eight studies to be high. Most studies had concerns related to blinding and allocation concealment.

Venepuncture compared to heel lance (with no sweet-tasting solution)

Pain scores

Venepuncture probably reduces pain compared to heel lance during the procedure when no sweet-tasting solution is administered (SMD -1.34 , 95% CI -1.55 to -1.14 ; $I^2 = 94\%$; 6 studies, 488 participants; moderate-certainty evidence).

Need for more than one skin puncture

Venepuncture probably reduces the need for more than one skin puncture compared to heel lance when no sweet tasting solution is administered (RR 0.63, 95% CI 0.40 to 0.97; $I^2 = 20\%$; 6 studies, 397 participants; moderate-certainty evidence)

Venepuncture compared to heel lance (with sweet-tasting solution)

Pain scores

Venepuncture may reduce pain compared to heel lance during the procedure when a sweet-tasting solution is administered (SMD -0.38 , 95% CI -0.69 to -0.07 ; $I^2 = 79\%$; 3 studies, 170 participants; low-certainty evidence).

Need for more than one skin puncture

The evidence is very uncertain about the effect of venepuncture compared to heel lance on the need for more than one skin puncture when a sweet-tasting solution is used (RR 7.00, 95% CI 0.38 to 129.93; I^2 not applicable; 2 studies, 110 participants; very low-certainty evidence). This is due to the small number of participants in each group.

Authors' conclusions

Based on moderate-certainty evidence, venepuncture probably reduces pain during the procedure and the need for more than one skin puncture compared to heel lance when no sweet-tasting solution is used. When sweet-tasting solutions are administered, venepuncture may reduce pain during the procedure compared to heel lance, based on low-certainty evidence. The evidence is very uncertain about the effect of venepuncture compared to heel lance on the number of skin punctures required in the presence of sweet-tasting solutions. This is due to the small sample size, and hence we could not draw any meaningful conclusions.

Further high-quality studies are needed, particularly in settings where sweet-tasting solutions are used, to better understand the comparative effectiveness of venepuncture and heel lance for neonatal blood sampling.

Funding

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Registration

Original review (2001): DOI 10.1002/14651858.CD001452

Review update (2004): DOI 10.1002/14651858.CD001452.pub2

Review update (2007): DOI 10.1002/14651858.CD001452.pub3

Review update (2011): DOI 10.1002/14651858.CD001452.pub4

PLAIN LANGUAGE SUMMARY

Is using a needle (venepuncture) better than using a device (heel lance) for taking a blood sample from full-term newborn babies?

Key messages

- Venepuncture (inserting a needle into a vein) may be less painful than heel lance (using a special device to puncture the skin) for newborn babies when having a blood sample taken, with or without sugar water for pain relief, and may reduce the need for more than one skin puncture when no sugar water is given for pain relief.

- We don't know which procedure, venepuncture or heel lance, is more painful for newborn babies when sugar is used for pain relief.
- We need better designed and larger studies that fully report their results to show which method is better for babies.

Why do we take blood samples from newborn babies?

In most countries, a blood sample from newborn babies is needed for routine screening tests. Blood samples may also be needed for other blood tests to diagnose and manage diseases, such as jaundice or low blood sugar (hypoglycemia). However, taking blood from babies is painful. Untreated pain in newborn babies may affect their physical growth and brain development. A sweet, sugar solution (such as sucrose) can be given to babies before the procedure, which helps to reduce the pain. It is difficult to measure pain in babies, but we rely on things like facial expressions, crying, and changes in heart rate to assess their pain.

How are blood samples taken?

Venepuncture

Venepuncture is a method of taking a blood sample where a person uses a needle to directly puncture the vein. The person who performs this must be well-trained to help ensure they successfully take a blood sample. This technique is often used when 1 mL or more of blood is required, and to prevent repeated punctures to the skin and associated complications (such as skin infections) from heel lance.

Heel lance

Heel lance is the standard way of taking a blood sample from newborn babies in hospitals worldwide. It is commonly used when the blood sample required is less than 1 mL. It does not require special training or expertise for successful use. The procedure involves making a small puncture (between 1 mm and 2 mm deep) on the heel of the foot using a lancet, which is usually an automatic device. Following the puncture, the infant's heel is squeezed to collect the required blood sample, which causes further pain.

What did we want to find out?

We wanted to find out:

- whether venepuncture is less painful than heel lance for newborn babies; and
- which procedure works best first time, so a second skin puncture is not needed.

What did we do?

We searched for studies that compared venepuncture and heel lance for taking blood samples in newborn babies. The studies needed to measure pain responses and the number of skin punctures needed for a successful sample. They also needed to record whether babies were given sugar solution as pain relief. Babies in the studies had to be healthy and born at full term. We compared and summarized the results of the studies and rated our confidence in the evidence, based on factors such as study methods and sizes.

What did we find?

We included eight studies involving 826 full-term newborn babies. Three studies took place in Sweden, and one each in Japan, Malaysia, Nepal, Spain, and the UK.

When no sweet-tasting solution was given:

- venepuncture probably causes less pain compared to heel lance (6 studies, 488 babies); and
- venepuncture probably reduces the need for more than one skin puncture compared to heel lance (6 studies, 397 babies).

When a sweet-tasting solution was given:

- venepuncture may cause less pain compared to heel lance (3 studies, 170 babies); and
- we are uncertain about the effect of venepuncture compared to heel lance on the need for more than one skin puncture (2 studies, 110 babies).

What are the limitations of the evidence?

Our confidence in the evidence was limited, and ranged from moderate to very low. This is because all the studies had limitations in how they were carried out. These included issues with how babies were assigned to each group, how the studies reported their results, and the number of babies included in the studies. In addition, the people doing the procedures knew which sampling method each baby received, which could have influenced how the studies assessed pain.

How up to date is this evidence?

This review updates our 2011 Cochrane review. The evidence is up to date to July 2025.

Venepuncture versus heel lance for blood sampling in term neonates (Review)

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SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Venepuncture compared to heel lance for term neonates for blood sampling, without sweet-tasting solution

Venepuncture compared to heel lance for term neonates for blood sampling, without sweet-tasting solution

Patient or population: term neonates for blood sampling, without sweet-tasting solution
Setting: neonatal units in high-income (Sweden, UK, Japan, Spain) and low-income (Nepal) countries
Intervention: venepuncture
Comparison: heel lance

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with heel lance	Risk with venepuncture				
Pain scores for venepuncture versus heel lance- total assessed with: NFCS: Neonatal Facial Coding System, NIPS: Neonatal Infant Pain scores, PIPP: Premature Infant Pain Profile	-	SMD 1.34 SD lower (1.55 lower to 1.14 lower)	-	488 (6 RCTs)	⊕⊕⊕⊖ Moderate ^{a,b}	Venepuncture probably reduces pain compared to heel lance during the procedure when no sweet-tasting solution is administered
Need for more than one skin puncture during blood sampling	209 per 1000	132 per 1000 (84 to 203)	RR 0.63 (0.40 to 0.97)	397 (6 RCTs)	⊕⊕⊕⊖ Moderate ^c	Venepuncture probably reduces the need for more than one skin puncture compared to heel lance

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_450393773805114930.

- ^a Downgraded one level for serious risk of bias due to concerns regarding: incomplete outcome reporting (one study), stopping the study early (one study), lack of blinding (three studies), and unclear randomization (two studies).
- ^b While there is substantial statistical heterogeneity ($I^2 = 94\%$), we did not downgrade for inconsistency because the direction of the effect was consistent across all studies (all favoring venepuncture over heel lance).
- ^c Downgraded one level for serious risk of bias due to: attrition bias (one study), stopping the study early (one study), unclear randomization (three studies), and lack of blinding (three studies).

Summary of findings 2. Summary of findings table - Venepuncture compared to heel lance for term neonates for blood sampling, who received a sweet-tasting solution

Venepuncture compared to heel lance for term neonates for blood sampling, who received a sweet-tasting solution

Patient or population: term neonates for blood sampling, who received a sweet-tasting solution

Setting: neonatal units in high-income (Japan) and upper-middle-income (Malaysia) countries

Intervention: venepuncture

Comparison: heel lance

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with heel lance	Risk with venepuncture				
Pain scores for venepuncture versus heel lance - total assessed with: NFCS: Neonatal Facial Coding System, PIPP: Premature Infant Pain Profile	-	SMD 0.38 lower (0.69 lower to 0.07 lower)	-	170 (3 RCTs)	⊕⊕⊕⊕ Low ^{a,b}	Venepuncture may reduce pain compared to heel lance during the procedure when a sweet-tasting solution is administered
Need for more than one skin puncture	0/55 events were observed in the heel lance group and 3/55 events were observed in the venepuncture group c			110 (2 RCTs)	⊕⊕⊕⊕ Very low ^{a,d}	The evidence is very uncertain about the effect of venepuncture compared to heel lance on the need for more than one skin puncture when a sweet-tasting solution is used

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_459868054425092856.

- ^a Downgraded one level for serious risk of bias due to concerns regarding: inappropriate random sequence generation (one study), use of a quasi-randomized design with no sequence generation (one study), and lack of blinding (one study).
- ^b Downgraded one level for serious inconsistency due to high heterogeneity ($I^2 = 79\%$), as the direction of effect was not consistent across all studies (one study).
- ^c The small number of overall study participants ($N = 110$) and the low number of event rates (3 events) suggest that the study was underpowered to detect a difference between the two procedures.
- ^d Downgraded two levels for extremely serious imprecision because the total number of neonates was below the optimal information size and the 95% CI were very wide.

BACKGROUND

Description of the condition

Relief of pain and suffering is a fundamental human right [1], and is considered important by clinicians, researchers, and parents [2, 3]. Although considered the “fifth vital sign” and despite the presence of numerous professional guidelines, neonates continue to experience iatrogenic pain (e.g. pain secondary to medical procedure or treatment) with no or minimal analgesia [4, 5, 6, 7, 8, 9, 10, 11, 12, 13]. Furthermore, unmanaged pain is associated with short-term and long-term complications, including physiological, behavioral, cognitive sequelae (e.g. memory issues, trouble concentrating, slower thinking), altered pain processing, impaired visual-motor integration, and poor executive functions (e.g. inability to plan, focus attention, manage time) [14, 15, 16, 17, 18, 19, 20]. Moreover, exposure to higher numbers of painful procedures is linked with poor cognition and motor outcomes at eight to 18 months [21, 22, 23], with magnetic resonance imaging (MRI) studies showing changes in white matter microstructure and smaller cerebellum in preterm infants ≤ 32 weeks [24, 25, 26]. Research also suggests that babies' early pain experiences may alter their pain response in later infancy [27, 28]. Pain management is therefore considered integral to good clinical practice.

A healthy full-term infant undergoes at least two painful procedures within the first 24 to 48 hours of life. Epidemiologic studies have demonstrated that infants in the neonatal intensive care unit (NICU) undergo an average of 14 to 16 painful procedures per day in the first two weeks after birth, most commonly with heel lance, venepuncture, venous cannulation, and peripheral arterial line [4, 5, 11, 13, 29, 30]. Traditionally, blood sampling of term neonates is performed using one of the two techniques, either heel lance or venepuncture. Both heel lance and venepuncture are considered to be moderately painful procedures [31]. Given this, non-pharmacological and pharmacological strategies are often administered to prevent or minimize procedural pain [32, 33, 34]. Despite various studies to date, there are no effective and practical methods to alleviate pain from heel lance or venepuncture [35].

Description of the intervention and how it might work

Venepuncture

Venepuncture is generally performed by a skilled practitioner who has training in phlebotomy. This technique is often used to collect more blood volume and to prevent repeated punctures and associated complications of heel lance, such as local cellulitis and skin necrosis [36]. Venepuncture is generally performed when 1 mL or more of blood volume is required, when a capillary sample is contraindicated (e.g. blood culture), or for certain laboratory tests (e.g. coagulation studies, or genetic and metabolic investigations). Other indications include administration of fluids for nutrition (e.g. glucose solution, total parenteral nutrition), and administration of medications or blood products.

Venepuncture is performed using an aseptic technique using a 21 or 23 G butterfly needle to obtain a blood sample, or using a 24 or 26 G intravenous catheter for administering intravenous fluids, medications, or blood products [36]. In infants in NICUs, the priority is to ensure that venous access sites are preserved as they may require prolonged intravenous fluid administration. Both procedures are generally performed with the use of pharmacological and non-pharmacological interventions

based on institutional preferences. The pain from venepuncture is mainly related to the needle puncture as, generally, no squeezing is required for blood collection. The advantages of venepuncture include a reduced risk of a hemolyzed or clotted blood sample, increased sample volume, and possibly less pain [37]. The disadvantage of venepuncture is the need for a skilled phlebotomist to perform the procedure. In contrast, the benefit of heel lance is the perceived ease of performing the task, even by paramedical personnel [36].

Heel lance

Heel lance is performed by a variety of healthcare professionals, ranging from physicians to nurses, and paramedical personnel, and special training is not required. Heel lance is performed when a capillary blood sample or the volume of a blood sample required is less than 1 mL [36, 38]. The procedure for heel lance begins with the healthcare professional washing their hands, and they then clean the medial or lateral aspect of the infant's heel with an alcohol/chlorhexidine swab. Once the skin has dried, they pierce the heel using an automated device to between 1 mm and 2 mm in depth, but no deeper than 2.4 mm [39]. Following the puncture, they squeeze the infant's heel to collect the required blood sample. Squeezing may contribute to additional pain. Once they obtain an adequate blood sample, they apply pressure using cotton wool or gauze to stop bleeding.

Both venepuncture and heel lance procedures are painful. Therefore, comparing standardized pain scores for term neonates undergoing either method as part of routine blood sampling would help to objectively compare which method is less painful, whilst assessing the success rate of each method.

Why it is important to do this review

Both venepuncture and heel lance can be used to obtain blood samples in neonates. The advantages of heel lance are that it can be performed by various healthcare professionals without specialized training. Given the necessity of frequent blood sampling in hospitalized neonates, it may be a more pragmatic procedure compared to venepuncture. Venepuncture, on the other hand, requires a skilled phlebotomist to perform the procedure. However, it has several advantages, such as obtaining adequate blood volume, reducing the risk of hemolyzed samples, and possibly causing less pain.

The aims of this review are to determine which procedure, venepuncture or heel lance, is less painful, and which procedure is more successful at obtaining an adequate blood sample at first attempt. This review is part of a suite of reviews by the Gerber Foundation addressing “The infant and family experience in the neonatal intensive care unit”. This is an update of a Cochrane review first published in 2001 and last updated in 2011 [40, 41].

OBJECTIVES

To evaluate the benefits (degree of pain) and harms (need for more than one skin puncture) of venepuncture compared to heel lance in term neonates.

METHODS

We conducted the review update following recommendations in the *Cochrane Handbook for Systematic Reviews of Interventions*

[42], and Methodological Expectations for Cochrane Intervention Reviews (MECIR) [43]. We reported the review's findings following PRISMA 2020 guidance [44]. In this review update, we amended the methods as follows to reflect current Cochrane standards.

- We created the summary of findings tables using the GRADE approach to assess the certainty of the evidence.
- Search strategies were edited to increase sensitivity and run without date limits. We omitted CINAHL as a search source because it is not requisite, as per the *Cochrane Handbook* [45]
- We updated the Plain Language Summary to reflect current Cochrane standards [46]
- Authorship changes: Dr Ohlsson is deceased. Gordon Martins, Michelle Fiander, and Anna Taddio joined the review author team.
- Objective: we clarified the wording of the objective to align with Cochrane standards.
- Study designs: we clarified that cluster RCTs were eligible for inclusion
- Interventions: we included the use of sucrose as a co-intervention, given the potential to influence pain outcomes.
- Outcomes: in the last published version of the Cochrane review [41], we included 11 secondary outcomes. In this review update, we include only one of these outcomes: "the need for more than one skin puncture". We removed the following outcomes as we deemed them either redundant (multiple cry outcomes), or not of primary importance to the practice of blood sampling, which always includes some pain.
 - Duration of first cry (seconds)
 - Total duration of cry (seconds)
 - Number of neonates who cried during the procedure
 - First crying time (seconds)
 - Total procedure time (seconds) (%)
 - Duration of cry (seconds) in the first three minutes after skin puncture
 - Sampling time (seconds)
 - Need for more than two skin punctures
 - Bruising/hematoma at a local site
 - Maternal anxiety score prior to the procedure
 - The infant's pain score as rated by the mother

Criteria for considering studies for this review

Types of studies

We included randomized controlled trials (RCTs), cluster-RCTs, and quasi-RCTs. We defined quasi-RCTs as those where trials allocated participants to different arms of the trial using a method of allocation that is not truly random, e.g. date of birth, medical record number, or date of recruitment.

We excluded non-randomized cohort studies. These are prone to bias due to confounding by indication or by residual confounding, both of which may influence the results of the studies.

Types of participants

We included healthy, term neonates ≥ 37 weeks' postmenstrual age (PMA) subjected to routine blood sampling and monitoring of hypoglycemia.

We excluded studies of preterm neonates, ≤ 37 weeks' PMA or sick term babies requiring NICU admission.

We included studies where participants extended beyond the gestational age cut-off, if $> 80\%$ of the neonates were term (≥ 37 weeks).

Types of interventions

We included studies where term neonates underwent blood sampling by venepuncture compared to heel lance. Neonates could receive a sweet-tasting solution, a commonly used analgesic in neonates, for pain relief [47, 48, 49, 50].

We excluded studies that assessed only one method of sampling, i.e. either venepuncture or heel lance.

Definitions

Venepuncture is a procedure in which a needle is used to take blood from a vein by directly puncturing the vein.

Heel lance is a method of blood collection which consists of making a puncture with a lancet (using an automated device to between 1 mm and 2 mm in depth, but no deeper than 2.4 mm) in one heel to obtain a blood sample.

Outcome measures

We included trials that measured the following outcomes irrespective of whether the trial(s) reported the data.

Critical outcomes

- Pain intensity scores were measured during the procedure by a validated composite pain assessment tool. Validated pain assessment tools included, but were not limited to, the following:
 - Neonatal Infant Pain Score (NIPS), scale 0 to 7; higher scores indicate greater pain [51];
 - Neonatal Facial Coding System (NFCS), scale 0 to 9; higher scores indicate greater pain [14, 52];
 - Premature Infant Pain Profile (PIPP), scale 0 to 21; higher scores indicate greater pain [53].

We excluded studies that reported on pain using physiological parameters in isolation as they are reflective of overall clinical status and non-specific arousal.

If a trial reported more than one of the pain scales specified above, we reported them both separately and combined. If a trial reported multiple pain scales at different time points for the same participant, we prioritized the score reported during the intervention.

Important outcomes

- Need for more than one skin puncture to collect blood during blood sampling.

Search methods for identification of studies

Electronic searches

The search strategies in the 2011 Cochrane review version were not documented in a reproducible manner [41]. Therefore, the Information Specialist (MF) on this review update wrote and

documented the new search strategies. The MEDLINE strategy was translated using appropriate controlled vocabulary and syntax for other databases. We limited the results by methodological filters to identify randomized clinical trials and systematic reviews. The source of the filters is stated in the search strategies. We conducted the searches without language restrictions. Searches for trials were run without date limits; searches for systematic reviews were limited to four years prior to the date of the search (04 July 2025). We have provided the full search strategies, preceded by a search narrative [54], in [Supplementary material 1](#).

We searched the following databases on 04 July 2025:

- Ovid MEDLINE® All, 1946 to Daily Update (04 July 2025);
- Ovid Embase, 1974 to 03 July 2025;
- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 07) in the Cochrane Library.

Searching other resources

Trial registries

We searched two clinical trial registries on 04 July 2025:

- ClinicalTrials.gov (<https://clinicaltrials.gov>);
- World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) (<https://trialsearch.who.int/Default.aspx>).

Conferences

We searched the following conference abstracts for the past five years (as available):

- Pediatric Academic Societies (PAS), 2020, 2022 to 2024;
- Perinatal Society of Australia and New Zealand (PSANZ), 2020 to 2024;
- European Academy of Paediatric Societies (EAPS), 2020 to 2023.

Other search methods

We searched the reference lists of trials included in this review update, and of systematic reviews related to the topic of this review. We searched for retractions of, or errata to, trials included in this review using the Retraction Watch Database (<https://retractiondatabase.org>) and PubMed. We consulted expert informants (pain researchers).

Data collection and analysis

Selection of studies

For screening and study selection, we used Covidence software [55]. We used Cochrane's Screen4Me (S4M), Known Assessments and RCT Classifier for 2023 search results [56, 57, 58, 59, 60, 61], and the RCT Classifier in Covidence for results of subsequent searches [55], to remove non-RCT records. We placed the references categorized as ineligible by S4M in the 'irrelevant' segment of Covidence [55]. This approach ensures references are available for deduplication purposes when searches are updated, and that references excluded by S4M are available for cross-checking should a potentially relevant study appear to have been missed by the search. Three review authors (GM, VS, or MF) independently screened the references that were remaining after S4M assessment and RCT Classifier processing. After assessment by title/abstract,

two review authors (GM and VS) independently assessed the full-texts of references assessed as potentially eligible. However, two review authors (GM and AT) assessed Shah 1997 [62], as the study was authored by review author VS. At any point in the screening process, we resolved disagreements through discussion.

For studies published in languages not read by the review authors, we sought translation by a native speaker to assess eligibility and extract key information (Alcaraz Sanz 1998 [63]).

We recorded the selection process in sufficient detail to complete a PRISMA flow diagram [44].

Data extraction and management

Three review authors (GM, VS, or AT) independently extracted data using a data extraction form that we designed. We piloted the form within the review team using a sample of two included studies. We resolved any disagreements by discussion. We identified and included one study by an author of this review (Shah 1997); two review authors (GM, AT) independently extracted data for this study and reached consensus through discussion.

In cases where we had questions about data or reporting, we emailed the study authors. We received no response to our request for clarification about data in one study (Shrestha 2012 [64]), so we estimated the mean and standard deviation (SD) values based on the article by Hozo and colleagues [65]. We successfully contacted and received clarifications and additional data from the authors of three studies (Eriksson 1999 [66]; Kvist 2002 [67]; Saththasivam 2009 [68]).

We extracted the following characteristics from each included study as available.

- Administrative details: study author(s) information; published or unpublished; year of publication; year in which study was conducted; presence of vested interest by study authors; and funding information ethics
- Study characteristics: details of randomization, study registration, study design type, study setting, number of study centers and location, and completeness of follow-up (e.g. greater than 80%)
- Participants: number randomized, number analyzed, mean gestational age, gestational age range, mean chronological age, chronological age range, birth weight, sex, state of arousal, reason for blood sampling, inclusion criteria and exclusion criteria
- Interventions: type and calibre of lancet used to perform heel lance, type and gauge of needle used to perform venepuncture, information on the operator who performed heel lance or venepuncture and the dose and concentration of sweet-tasting solution administered (co-intervention)
- Outcomes as described in the [Outcome measures](#) section

We described ongoing studies and documented available information (such as the primary author, research question(s), methods, outcome measures, and the estimated reporting date) in [Supplementary material 4](#).

Two review authors (GM, VS) used Review Manager (RevMan) for data entry [69]. We replaced any standard error of the mean (SEM) with the corresponding SD.

Risk of bias assessment in included studies

Three review authors (GM and VS; or GM and AT for the Shah 1997 study) independently assessed the risk of bias (low, high, or unclear) of all included trials using the Cochrane Risk of bias tool 1 (RoB 1) [70], for the following domains.

- Sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment (detection bias)
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Any other bias

We resolved any disagreements by discussion or by consulting a third review author (AT). For Shah 1997, the two review authors (GM, AT) independently assessed the risk of bias for this study and reached consensus through discussion. See [Supplementary material 7](#) for a more detailed description of the risk of bias for each domain.

Measures of treatment effect

For dichotomous outcomes, we used risk ratios (RRs) with corresponding 95% confidence intervals (CIs) to summarize treatment effects across studies. We selected RR as the effect measure because it is easily interpretable and commonly used in neonatal intervention studies.

For continuous outcomes, we used the standardized mean difference (SMD) with 95% CIs, as included studies used different scales or pain scores to assess the same outcome domain (e.g. neonatal pain). The use of SMD allowed us to synthesize data across studies using different instruments. Where possible, we interpreted effect sizes using conventional thresholds (e.g. 0.2 = small, 0.5 = moderate, 0.8 = large), although we acknowledge the limitations of such arbitrary thresholds in clinical interpretation.

Where studies reported medians and ranges rather than means and SDs, we converted these to means and SDs using the method proposed by Hozo and colleagues [65].

When meta-analysis was not feasible or appropriate due to heterogeneity in outcome measures or lack of comparable data, we intended to synthesize findings narratively. Where possible, we planned to use vote counting based on the direction of effect and generate effect direction plots to summarize results across studies, following the recommendations of Campbell and colleagues [71]. This approach provides a structured method to interpret heterogeneous data when statistical pooling is not possible.

Unit of analysis issues

The unit of analysis was the participating infant in individually randomized trials, and we considered each infant only once in an analysis. For trials with more than two treatment groups (Eriksson 1999; Ogawa 2005 [72]), we identified distinct pairs of interventions and controls. Since these trials compared venepuncture (with and without sweet-tasting solutions) against corresponding heel lance groups, we included each infant in only one comparison. Therefore, the number of participants for each of the four groups is independent, and there was no double-counting.

Had we identified cluster-RCTs, we would have included them in the analysis, provided appropriate statistical methods were used, or sufficient data were available to adjust for clustering. We planned to consider the participating neonatal (or hospital) unit as the unit of analysis and to analyze data using an estimate of the intracluster correlation coefficient (ICC) derived from the trial (if possible) or from a similar trial or from a study with a similar population, as described in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* [73]. If the ICC was not reported, we would have attempted to obtain it from the study authors or used estimates from similar studies. Where adjustment was not possible, we would have reported the data narratively and not pooled them with individually randomized trials.

Dealing with missing data

For all the included studies, we noted the levels of attrition. We planned to perform sensitivity analyses to explore the impact of including studies with high levels of missing data in the overall assessment of treatment effect. For all outcomes, we carried out analyses, as far as possible, on an intention-to-treat (ITT) basis. Whenever possible, we analyzed all participants in the treatment group to which they were randomized, regardless of the actual treatment received. The denominator for each outcome in each trial was the number randomized minus any participants whose outcomes were known to be missing.

Reporting bias assessment

Due to the small number of studies included in each meta-analysis (< 10 studies), we did not conduct formal assessments of reporting bias (such as publication bias) using funnel plots or statistical tests for funnel plot asymmetry. These methods are unreliable with a small number of studies. As such, we were unable to rule out the possibility of reporting bias or small study effects.

Synthesis methods

We considered studies eligible for inclusion in each synthesis based on similarities in population characteristics, interventions, comparators, and outcomes, as prespecified in our inclusion criteria. To determine compatibility for synthesis, we compared study characteristics using structured tables that detailed intervention components, outcome definitions, and time points of measurement.

When the studies were sufficiently homogeneous in terms of participants, interventions, and outcome measurement, we performed a meta-analysis using RevMan [69]. For dichotomous outcomes, we calculated RRs with corresponding 95% CIs. For continuous outcomes, we used the SMD with 95% CIs, given the use of different measurement scales across studies.

We used a fixed-effect model when we judged clinical and methodological characteristics to be sufficiently similar across studies, and we considered the assumption of a common treatment effect plausible. We acknowledge that fixed-effect meta-analysis assumes that observed differences in treatment effect across studies are due to chance alone.

Had we been unable to perform meta-analysis (due to, for instance, substantial clinical heterogeneity or insufficient data), we would have undertaken synthesis without meta-analysis (SWiM) [71]. We would have grouped results by outcome and intervention type,

described the direction and magnitude of effects, and discussed any patterns observed across studies.

We made all synthesis decisions with consideration of clinical relevance and consistency across studies, and aimed to limit the number of comparisons to those that directly addressed our review objectives.

Investigation of heterogeneity and subgroup analysis

We assessed heterogeneity using the χ^2 test and quantified inconsistencies with the I^2 statistic, and interpreted I^2 values based on the following thresholds recommended in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [74].

- < 25%: no heterogeneity
- 25 to 49%: low heterogeneity
- 50 to 74%: moderate heterogeneity
- $\geq$ 75%: high heterogeneity

We planned to investigate potential sources of clinical and methodological diversity through subgroup analyses when at least 10 studies were available per outcome. The following predefined subgroup comparisons were planned.

- Use of sweet-tasting solutions versus no sweet-tasting solution, based on evidence that oral sucrose or glucose may modulate pain perception in neonates
- Gestational age: only term infants, to examine whether findings differed from studies including preterm populations
- Study setting: studies conducted only in the postnatal ward, to distinguish effects in routine care from those in intensive care environments
- Pain assessment tool used: NIPS, PIPP, or NFCS, given possible variation in sensitivity and scoring across scales

We planned these subgroup analyses using study-level characteristics, with each study categorized into only one level of each subgroup factor. If sufficient data were available, we planned to use tests for interaction to assess whether effects differed statistically between subgroups.

Due to the limited number of studies included for each outcome (≤ 6 studies), we could not perform formal subgroup analyses. However, we explored clinical heterogeneity narratively, and considered the differences in populations, interventions, and outcome measurement tools when interpreting the findings.

Equity-related assessment

We did not investigate health inequities in this Cochrane review.

Sensitivity analysis

We performed sensitivity analyses to assess the robustness of the results as some studies did not provide mean and SD values, which we had to estimate. Since there is no formal statistical test for sensitivity analysis, we used informal comparisons to determine whether the treatment effects changed under different assumptions.

Specifically, we tested the robustness of our critical outcome, pain scores, by excluding studies where we estimated means and SDs by converting from medians and ranges (e.g. Larsson 1998 [75, 76];

Ogawa 2005; Shrestha 2012). We did not use changes in P values to judge these differences, acknowledging that statistical significance may be lost when fewer studies are included. Given that there is no formal statistical test that can be used for sensitivity analysis, we planned to provide informal comparisons between the different ways of estimating the effect under different assumptions. Changes in the P values should not be used to judge whether there is a difference between the main analysis and sensitivity analysis, since statistical significance may be lost with fewer studies included.

We planned to report sensitivity analysis results in tables, rather than forest plots.

Where we identified substantial heterogeneity, we planned to conduct sensitivity analysis to determine if the findings were affected by the inclusion of only those trials considered to have used adequate methodology, i.e. those with a low risk of bias. We planned to report the results of sensitivity analyses for the critical outcome only. However, we did not perform any sensitivity analysis, as it was not required.

Certainty of the evidence assessment

We used the GRADE approach, as outlined in the GRADE Handbook [77], to assess the certainty of evidence of the following (clinically relevant) outcomes.

- Pain response as measured by a validated pain score
- Need for more than one skin puncture during the sampling session

Two review authors (GM and VS) independently assessed the certainty of the evidence for all studies except Shah 1997, which GM and AT assessed. We resolved disagreements by discussion. We considered evidence from RCTs as high certainty, but downgraded the evidence one level for serious (or two levels for very serious) limitations based upon the following: design (risk of bias), consistency across studies, directness of the evidence, precision of estimates, and presence of publication bias. We used the GRADEpro Guideline Development Tool to create two summary of findings tables [78], and reported the certainty of the evidence with or without the following non-pharmacological interventions for the following comparisons:

- Comparison 1: Venepuncture compared to heel lance in neonates who did not receive a sweet-tasting solution; and
- Comparison 2: Venepuncture compared to heel lance in neonates who received a sweet-tasting solution.

The GRADE approach results in an assessment of the certainty of a body of evidence as one of four grades.

- High certainty: further research is very unlikely to change our confidence in the estimate of effect.
- Moderate certainty: further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.
- Low certainty: further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.
- Very low certainty: we are very uncertain about the estimate.

We justified all decisions to downgrade the certainty of the evidence using footnotes, and made comments to aid the reader's understanding of the review where necessary.

Consumer involvement

We did not involve consumers in this review update due to limited resources.

RESULTS

Description of studies

For details, see [Supplementary material 2](#), [Supplementary material 3](#), and [Supplementary material 4](#).

Results of the search

Searches identified 12,751 references; after removing 9171 duplicates, 3580 references were available for screening. We used Cochrane's Screen4Me, RCT Classifier and Known Assessments, to assess results of the 2023 search; and the Covidence RCT Classifier [55] to identify non-RCTs for 2024 and 2025 searches.

We excluded 3566 based on title abstract (519 via Screen4Me; 13 via the Covidence RCT Classifier [55]; and 3034 by authors). We reviewed 14 full texts or trial registry records. We excluded four studies (four references); identified one ongoing study (one reference); and included eight studies (nine references). Two studies are new to this review update version.

Study selection details are available in [Figure 1](#). Details of the Screen4Me workflow are in [Figure 2](#).

Figure 1.

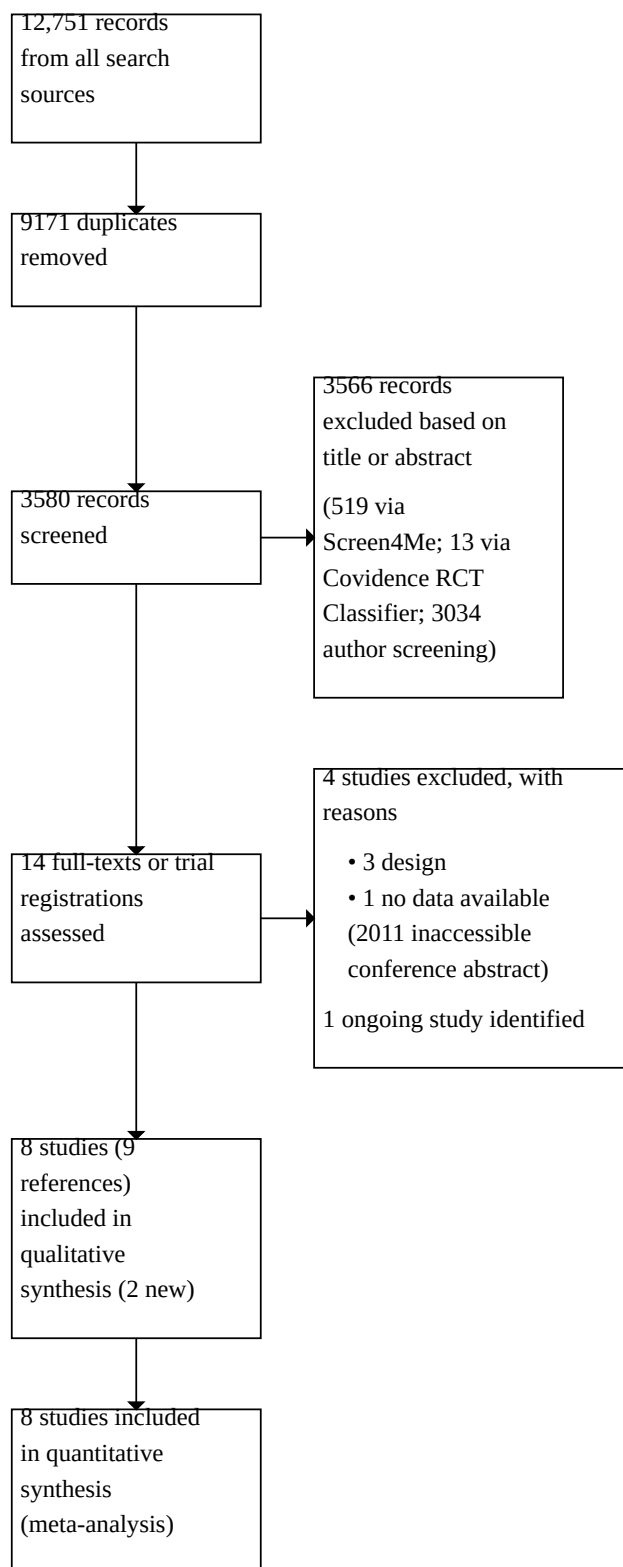
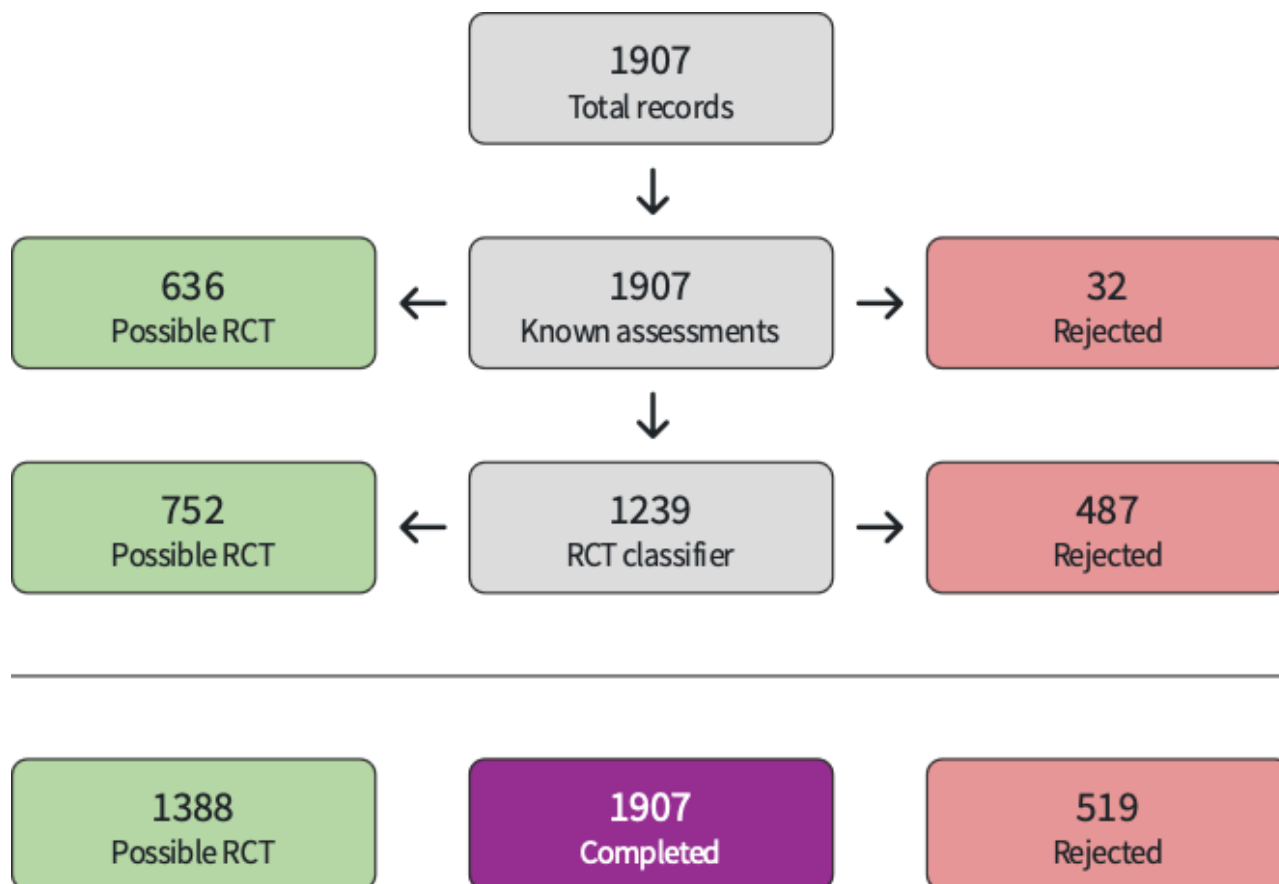


Figure 1. (Continued)



Figure 2. Cochrane's Screen4Me workflow results



Included studies

We included eight RCTs involving a total of 826 neonates, of which data from 767 neonates were eligible for this review update. We identified two new studies in this review update (Alcaraz Sanz 1998; Shrestha 2012). See [Table 1](#) for a summary of key study characteristics, and [Supplementary material 2](#) for detailed trial-level data.

Location

The trials were conducted in six countries: Japan (Ogawa 2005), Malaysia (Saththasivam 2009), Nepal (Shrestha 2012), Spain (Alcaraz Sanz 1998), Sweden (Eriksson 1999, Kvist 2002, Larsson 1998), and the UK (Shah 1997).

Population

All studies recruited neonates, primarily term infants (PMA > 37 weeks). One study included a small number of late preterm infants (PMA 36 to 43 weeks; median 40 weeks), and was included as over

80% of the infants were term (Larsson 1998). Study settings varied, and included postnatal wards and hospital-based settings.

Interventions

All eight included studies compared heel lance with venepuncture as blood sampling techniques. In three studies, a sweet-tasting solution (such as glucose or sucrose) was administered to all or some participants prior to the procedure (Eriksson 1999; Ogawa 2005; Saththasivam 2009).

Outcomes

All studies reported pain as the critical outcome. Studies assessed pain using the following validated tools.

- NIPS (Kvist 2002; Shah 1997; Shrestha 2012)
- PIPP (Eriksson 1999)
- NFCS (Larsson 1998; Ogawa 2005; Saththasivam 2009)
- Behavioral cry measures (Alcaraz Sanz 1998)

Studies reported pain scores using a mix of statistical formats (means, medians, percentages), which affected inclusion in meta-analysis. Shrestha 2012 and Ogawa 2005 reported outcomes as medians and interquartile ranges or graphical centiles. For Saththasivam 2009, the authors provided mean and SD data upon request. For Shrestha 2012, we did not receive any response from the study authors. We generated estimates using the method described by Hozo and colleagues to convert medians and ranges [65].

Ongoing studies

We identified one ongoing study that compared NFCS scores in venepuncture in hand and heel lance in newborns (RBR-2wkqpw [79]). See [Supplementary material 4](#) for details.

Studies awaiting classification

We did not identify any studies that are awaiting classification.

Excluded studies

We excluded four studies. Three were not RCTs (Correcher 2012 [80]; Logan 1999 [81]; Shah 2006 [82]). One was an unpublished conference abstract, for which we had only the title (Bomben 2011 [83]). We have provided further details in [Supplementary material 3](#).

Risk of bias in included studies

See [Figure 3](#), [Figure 4](#), and [Supplementary material 2](#) for details.

Figure 3. Risk of bias graph

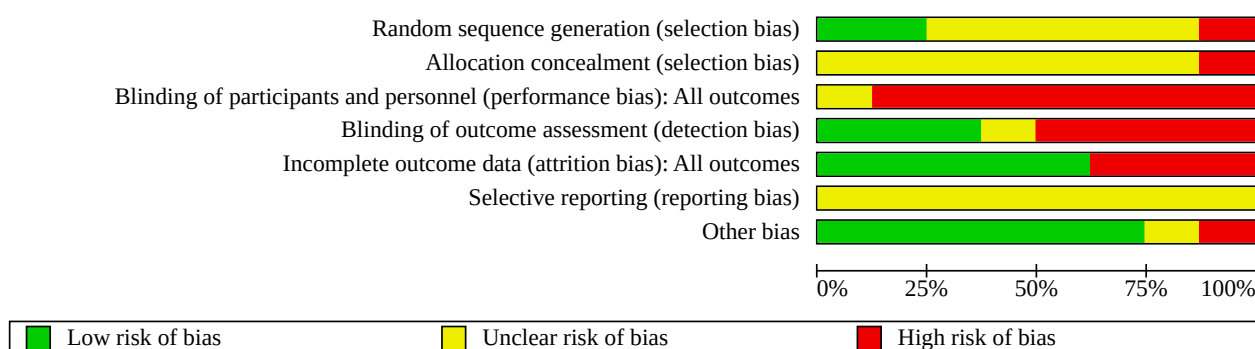


Figure 4. Risk of bias summary

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Alcaraz Sanz 1998	?	?	-	?	+	?	?
Eriksson 1999	?	?	-	-	+	?	+
Kvist 2002	?	?	-	-	-	?	+
Larsson 1998	?	?	-	+	-	?	+
Ogawa 2005	?	?	-	+	+	?	+
Saththasivam 2009	-	-	-	+	-	?	+
Shah 1997	+	?	?	-	+	?	-
Shrestha 2012	+	?	-	-	+	?	+

The risk of bias across the included studies varied. Several trials showed methodological concerns in one or more domains. Many studies lacked detailed reporting, thus limiting our ability to make robust assessments. The overall risk of bias in the eight studies was judged to be high.

Random sequence generation

We assessed two studies (25%) at low risk of bias, one study (12.5%) at high risk of bias, and five studies (62.5%) at unclear risk of bias.

Allocation concealment

We assessed seven studies (87.5%) at unclear risk of bias, and one study (12.5%) at high risk of bias.

Blinding

Blinding of intervention was generally not feasible due to the nature of the procedures.

- Detection bias: we assessed three studies (37.5%) of having low risk of bias, four studies (50%) of having high risk of bias, and one study (12.5%) of having unclear risk of bias.
- Performance bias: there was a high risk of bias in the blinding of participants and personnel in seven studies (87.5%), and one study (12.5%) had an unclear risk of bias.

Incomplete outcome data

- Five studies (62.5%) reported outcomes for all randomized participants, and we assessed them as low risk of bias.
- We assessed one study (12.5%) as at high risk of bias, as 30 infants from the venepuncture group using a large calibre needle were excluded as the procedure was unsuccessful in 10. The study authors did not provide data on the NIPS score or number of skin punctures, due to the small sample size in that group (Kvist 2002).
- We judged one study (12.5%) to have a high risk of bias as six infants were excluded, as these infants had cried prior to heel lance or venepuncture, or both (Saththasivam 2009)

- We assessed one study (12.5%) as at high risk of bias as three infants in the heel lance group were excluded as two infants cried prior to the procedure and one infant was < 35 weeks gestation. Twenty infants enrolled in the heel lance group that used a large lancet were excluded due to unequal enrolment compared to the other two groups (heel lance using a small lancet and venepuncture group) (Larsson 1998).

Selective reporting

- No trial protocols were publicly available, so we were unable to fully assess selective reporting in the included studies. Hence, we assessed all eight studies (100%) to have unclear risk of bias.
- However, all included studies appeared to report on their critical outcomes, as described in their methods.

Other potential sources of bias

We judged that the risk of 'other' bias was unclear in one study (12.5%), high risk of bias in one study (12.5%), and low risk of bias in six studies (75%).

Synthesis of results

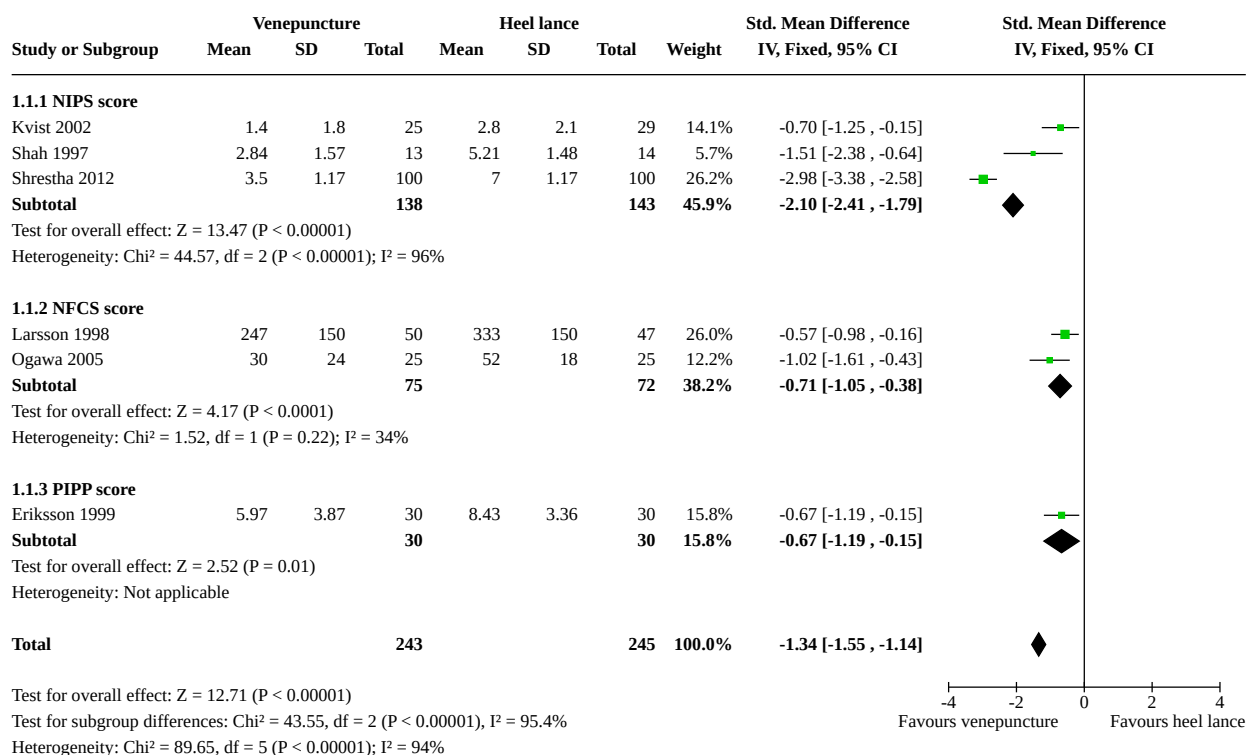
Comparison 1: Venepuncture compared to heel lance without a sweet-tasting solution

See [Summary of findings 1](#).

Critical outcomes

Pain response (measured using one or more of NIPS, NFCS and/or PIPP, during the procedure)

Six studies reported this outcome (Eriksson 1999; Kvist 2002; Larsson 1998; Ogawa 2005; Shah 1997; Shrestha 2012). Venepuncture probably reduces pain compared to heel lance when no sweet-tasting solution is administered (SMD -1.34, 95% CI -1.55 to -1.14; $I^2 = 94\%$; 6 studies, 488 participants; moderate-certainty evidence; [Figure 5](#)).

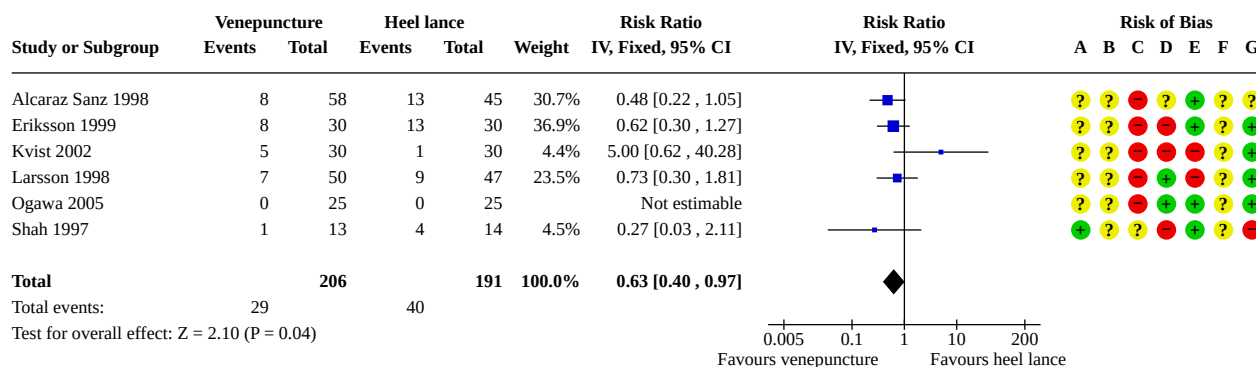
Figure 5. Forest plot of comparison: 1 Outcomes for venepuncture versus heel lance in neonates who did not receive a sweet-tasting solution: 1.1 Pain scores for venepuncture versus heel lance.

Important outcomes

Need for more than one skin puncture during the procedure

Six studies reported this outcome (Alcaraz Sanz 1998; Eriksson 1999; Kvist 2002; Ogawa 2005; Larsson 1998; Shah 1997).

Venepuncture probably reduces the need for more than one skin puncture compared to heel lance (RR 0.63, 95% CI 0.40 to 0.97; $I^2 = 20\%$; 6 studies, 397 participants; moderate-certainty evidence; [Figure 6](#)).

Figure 6. Forest plot of comparison: 1 Outcomes for venepuncture versus heel lance in neonates who did not receive sweet-tasting solution: 1.2 Need for more than one skin puncture for venepuncture versus heel lance.**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

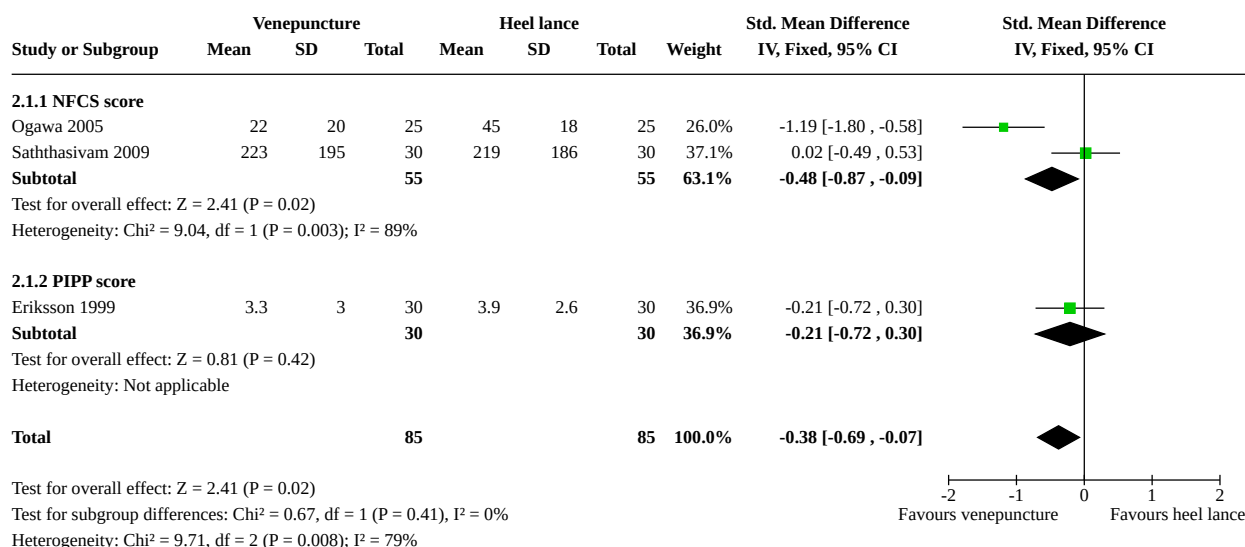
Comparison 2: Venepuncture compared to heel lance with a sweet-tasting solution

See [Summary of findings 2](#).

Critical outcomes

Pain response (measured using one or more of NIPS, NFCS and/or PIPP, during the procedure)

Three studies reported this outcome (Eriksson 1999; Ogawa 2005; Saththasivam 2009). Venepuncture may reduce pain compared to heel lance when a sweet-tasting solution is administered (SMD -0.38, 95% CI -0.69 to -0.07; I² = 79%; 3 studies, 170 participants; low-certainty evidence; [Figure 7](#)).

Figure 7. Forest plot of comparison: 2 Outcomes for venepuncture versus heel lance in neonates who received a sweet-tasting solution: 2.1 Pain scores for venepuncture versus heel lance.

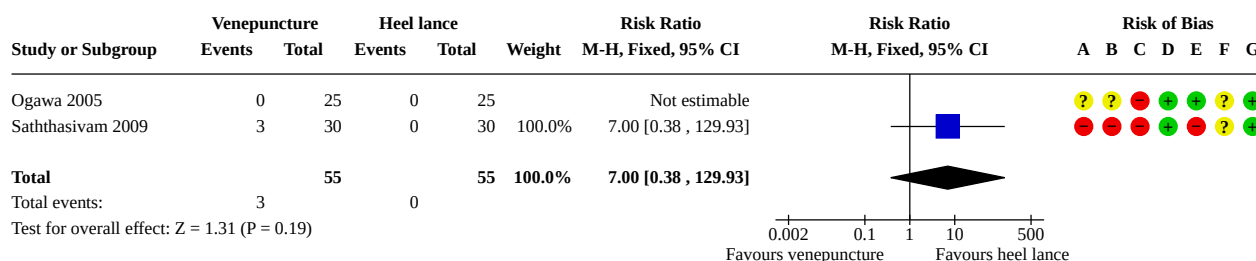
Important outcomes

Need for more than one skin puncture during the procedure

Two studies reported this outcome (Ogawa 2005; Saththasivam 2009). The evidence is very uncertain about the effect of

venepuncture compared to heel lance on the need for repeat skin puncture in the context of sweet-tasting solutions (RR 7.00, 95% CI 0.38 to 129.93; I^2 not applicable; 2 studies, 110 participants; very low-certainty evidence; [Figure 8](#)).

Figure 8. Forest plot of comparison: 1 Outcomes for venepuncture versus heel lance in neonates who received a sweet-tasting solution: 2.2 Need for more than one skin puncture for venepuncture versus heel lance.



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Sensitivity analysis

We performed sensitivity analyses for our critical outcomes of pain scores, to assess the robustness of our conclusions. When we excluded trials that used data estimation from graphical reports or medians, the direction and statistical significance of the effects remained unchanged. This confirms that the choice of data extraction method did not essentially alter the main findings of this review.

Reporting biases

We assessed all eight included studies as having an unclear risk of reporting bias. None of the trials were prospectively registered in a trial registry, and study protocols were unavailable. Thus, we were unable to determine whether the included studies reported all prespecified outcomes, which raises the possibility of selective outcome reporting.

Due to the small number of studies included in each meta-analysis (< 10 studies), we could not conduct formal assessments of reporting bias (such as publication bias) using funnel plots or statistical tests for funnel plot asymmetry. These methods are unreliable with only a small number of included studies. As such, we were unable to rule out the possibility of reporting bias or small study effects.

If future updates of this review include 10 or more studies in any meta-analysis, we plan to assess the risk of bias due to missing results using visual inspection of funnel plots and statistical tests for funnel plot asymmetry. For dichotomous outcomes, we plan to use the test proposed by Harbord and colleagues [84]; for continuous outcomes, we will use the method proposed by Egger and colleagues [85]. Two review authors (VS and GM) will independently

assess reporting bias, and will resolve disagreements through discussion or consulting a third review author (AT).

We contacted study authors where we required clarification or missing information. Three study authors provided us with additional information and clarification (Eriksson 1999; Kvist 2002; Saththasivam 2009).

DISCUSSION

Summary of main results

We included eight studies comparing venepuncture to heel lance for blood sampling in term neonates, with sample sizes ranging from 27 to 200 infants. In this review update, we included two new studies (Alcaraz Sanz 1998; Shrestha 2012), and identified one ongoing study (RBR-2wkqpw). Our analyses found the following.

- Compared to heel lance, venepuncture was associated with reduced pain responses as assessed by validated pain scores (NIPS, NFCS, or PIPP), both in studies with and without the use of sweet-tasting solutions.
- Venepuncture was also associated with a lower need for repeat sampling, suggesting greater first-attempt success compared to heel lance, without sweet-tasting solutions. The evidence was inconclusive for sampling success with either method, with the administration of the sweet-tasting solution.
- The certainty of evidence is limited by several factors, including incomplete outcome reporting in some studies and the relatively small number of participants in some analyses.

To address these limitations, future high-quality RCTs with standardized methodologies and adequate sample sizes should be

conducted in routine postnatal settings to confirm these findings and improve generalizability.

Limitations of the evidence included in the review

The included studies were conducted across a diverse range of settings in six countries (Japan, Malaysia, Nepal, Spain, Sweden, and the UK), primarily involving term neonates in both postnatal wards and other hospital-based settings. One study included a small number of late preterm infants. However, as the median gestational age was 40 weeks, we included data from the study (Larsson 1998).

While the included studies addressed the primary review question—comparing venepuncture to heel lance for blood sampling in neonates—the evidence base had several limitations. Study sample sizes were small (ranging from 27 to 200), limiting the power of individual studies and contributing to imprecision in effect estimates. The included studies assessed pain, the critical outcome across all included studies, using a range of validated tools (NIPS, PIPP and NFCS). However, the variability in outcome measurement, statistical reporting formats (means, medians, percentages), and incomplete reporting limited comparability and inclusion of studies in meta-analyses. Two studies reported the outcomes either as medians with interquartile ranges or graphically, requiring estimation methods to convert data for analysis (Ogawa 2005; Shrestha 2012). For one study, Saththasivam 2009, the study authors provided summary statistics upon request. However, we did not receive a response for another study (Shrestha 2012).

Heterogeneity in procedural factors further limited interpretability. Operator experience and number varied widely across studies and were not consistently reported. This variability likely contributed to statistical heterogeneity and reflects a real-world challenge in standardizing procedural techniques in neonatal settings. Additionally, three studies administered sweet-tasting solutions prior to the procedure, potentially modifying pain responses (Eriksson 1999; Ogawa 2005; Saththasivam 2009).

Due to these factors, we downgraded the certainty of the evidence in several domains, particularly for risk of bias (due to unclear randomization and blinding procedures in some studies), inconsistency (statistical heterogeneity), and imprecision (small sample sizes). Despite these limitations, the studies did broadly represent the intended population and interventions relevant to clinical practice [86].

Limitations of the review processes

A limitation of the review process was our inability to obtain clarifications or additional data from some corresponding trial authors, despite our attempts to contact them. This may have impacted the completeness and accuracy of data extraction for certain outcomes, particularly where results were reported using non-standard formats or lacked measures of dispersion. As a result, we made assumptions in some cases (e.g. conversion of medians to means), which could introduce bias or imprecision in the synthesis. Additionally, our inability to access missing data limited the ability to fully assess the risk of bias or conduct subgroup analyses in some studies. While we used comprehensive search strategies, and performed screening and extraction in duplicate, the potential for unreported or inaccessible study data remains a limitation.

Agreements and disagreements with other studies or reviews

To our knowledge, there are no other published systematic reviews directly comparing venepuncture to heel lance for blood sampling in neonates. As such, we were unable to compare our findings with other comprehensive reviews. The findings of this Cochrane review update therefore contribute novel evidence to the existing literature, and help fill a gap in current understanding of pain outcomes associated with these common procedures.

AUTHORS' CONCLUSIONS

Implications for practice

This review identified eight randomized controlled trials (RCTs) involving 826 neonates that compared venepuncture with heel lance for blood sampling. The available evidence suggests that venepuncture is associated with lower levels of pain, even with sweet-tasting solution as a co-intervention, and a lower need for repeat sampling compared to heel lance when no sweet tasting solution is given. However, the certainty of the evidence is moderate to low, primarily due to methodological limitations, inconsistency across studies, and imprecision. The evidence is very uncertain about the effect of venepuncture versus heel lance on the need for more than one skin puncture when a sweet-tasting solution is used.

The potential for venepuncture to cause less pain than heel lance may be relevant when considering approaches to minimize procedural pain in neonates. However, the selection of a blood sampling technique may also depend on other factors, including the clinical setting, operator training and experience, availability of resources, and the neonate's clinical status.

As the evidence is uncertain, decisions about the preferred method for blood sampling in neonates should take into account the likely balance between potential benefits and harms, as well as contextual factors such as staff expertise and institutional protocols. Further research is needed to strengthen the evidence base before any definitive changes in clinical practice or policy can be supported.

Implications for research

The findings of this Cochrane review update highlight several important gaps in the current evidence base comparing venepuncture to heel lance for blood sampling in neonates.

Future research should prioritize well-designed, adequately-powered RCTs that specifically address these limitations. Studies should ensure clear reporting of methodology, including randomization procedures, allocation concealment, and blinding where possible. Study investigators should use validated pain assessment tools and provide complete outcome data in formats that allow for meaningful data synthesis. Standardization in outcome measures and timing of assessments would improve comparability between studies.

Further trials should also examine the influence of gestational age and clinical setting on the effectiveness and tolerability of each sampling method. Inclusion of both term and preterm infants, as well as diversity in healthcare settings, would enhance the applicability of findings. Detailed reporting of operator training

and experience is necessary to evaluate the potential impact of technical expertise on outcomes.

As pain is a critical outcome, future studies should consider both immediate and short-term pain responses, along with the need for repeat sampling. Safety outcomes, including bruising, infection, or other procedure-related complications, should also be systematically assessed and reported.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD001452.pub5](https://doi.org/10.1002/14651858.CD001452.pub5).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 Risk of bias tool

ADDITIONAL INFORMATION

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We would like to acknowledge the intellectual contributions of Dr Arne Ohlsson to the previously published versions of this Cochrane review [40, 41, 87, 88].

We did not use any artificial intelligence tools in this review.

Editorial and peer-reviewer contributions

Cochrane Neonatal supported the authors in the development of this Cochrane intervention review update.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Anju Sinha, Indian Council of Medical Research, Division of Child Health, Ansari Nagar, New Delhi;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Alan Thomas, Cochrane Central Editorial Service;

- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Deirdre Walshe, Cochrane Central Production Service;
- Peer reviewers (provided comments and recommended an editorial decision):
 - Radheshyam Meher, All India Institute of Medical Sciences, New Delhi, India (patient and public review);
 - Clare Miles, Evidence Production and Methods Directorate (methods review);
 - Jo Platt, Central Editorial Information Specialist (search review).

Contributions of authors

GM and VS selected studies, extracted data, assessed risk of bias, conducted GRADE assessments, and wrote the review.

AT and GM conducted the risk of bias assessments, data extraction, and GRADE for Shah 1997.

MF wrote the search strategies, search methods, results of search, and PRISMA; checked narrative data elements in characteristics of studies; and wrote the review.

Dr Ohlsson (deceased) was involved in previous versions of this Cochrane review [40, 41]. Some of the content retained in this review reflects his contribution.

Declarations of interest

Dr Vibhuti Shah is the principal author of one of the trials included in this review (Shah 1997). She did not assess eligibility, extract data, or conduct risk of bias or GRADE assessments for this study. GM and AT undertook this work for this study. Dr Shah is an Associate Editor for Cochrane Neonatal, but did not participate in the acceptance or editorial processes for this review.

Michelle Fiander is a Managing Editor and Information Specialist for Cochrane Neonatal Group, but did not participate in the editorial processing or acceptance of this review.

Dr Gordon Martins is a Neonatology fellow and has no interests to declare.

Anna Taddio is a co-author and is the GSK Chair in Vaccine Education and Practice-Oriented Tools (funding through endowment to the university). The University of Toronto holds a Section 9 Trademark No. 924835 for CARD™, a knowledge translation tool for reducing distress during vaccination; there is no associated financial gain. AT has received honoraria for participation in the Vaccine Action Plan Advisory Group, Ontario Pharmacists Association.

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Registration and protocol

Original review (2001): DOI 10.1002/14651858.CD001452

Review update (2004): DOI 10.1002/14651858.CD001452.pub2

What's new

Review update (2007): DOI 10.1002/14651858.CD001452.pub3

Review update (2011): DOI 10.1002/14651858.CD001452.pub4

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (see [Supplementary material 6](#)): full search strategies for each database ([Supplementary material 1](#)); full citations of each unique report for all studies included ([Supplementary material 2](#)), excluded at the full-text screen ([Supplementary material 3](#)), or awaiting classification ([Supplementary material 4](#)) in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows ([Supplementary material 5](#)). Appropriate permissions have been obtained for such use. We conducted analyses and data management within Cochrane’s authoring tool, RevMan [69], using the inbuilt computation methods. Template data extraction forms (for example, from Covidence [55], and Excel) are available from the review authors on reasonable request.

Date	Event	Description
17 July 2026	New citation required and conclusions have changed	We included two additional studies: Alcaraz Sanz 1998 and Shrestha 2012. If no sweet-tasting solution is used, venepuncture probably reduces pain during the procedure and the need for more than one skin puncture compared to heel lance. Venepuncture may reduce pain during the procedure compared to heel lance when sweet-tasting solution is used. When sweet-tasting solution is used, the evidence for the need for more than one skin puncture is very uncertain.
17 July 2026	New search has been performed	New search performed 04 July 2025. We updated the methods to reflect current Cochrane standards. Authorship changes: Gordon Martins, Michelle Fiander, and Anna Taddio joined the author team.

History

Protocol first published: Issue 2, 1999

Review first published: Issue 2, 1999

Date	Event	Description
27 January 2020	Amended	Arne Ohlsson deceased.
7 December 2011	Amended	Minor amendment to correct Outcome numbering in text, 'Effects of Interventions', Comparison 2.
8 July 2011	New citation required and conclusions have changed	For this update we included comparisons for studies that did or did not include a sweet tasting solution as a co-intervention for venepuncture and heel lance. As before, venepuncture proved to be less painful than heel lance, but the addition of a sweet tasting solution prior to skin

Date	Event	Description
		puncture reduced the difference between the two methods used for blood sampling.
8 July 2011	New search has been performed	This updates the review 'Venepuncture versus heel lance for blood sampling in term neonates' published in the Cochrane Database of Systematic Reviews [41]. Search updated May 12, 2011. One additional trial was identified.
11 June 2008	Amended	Converted to new review format.
27 July 2007	New citation required but conclusions have not changed	Substantive amendment
27 July 2007	New search has been performed	This updates the review "Venepuncture versus heel lance for blood sampling in term neonates", published in The Cochrane Library, Issue 2, 1999 [40] and updated in 2011 [41]. One additional randomized controlled trial was identified for inclusion. The data from this additional study strengthen the evidence that venepuncture is less painful compared to heel lance to obtain a blood sample from healthy term neonates. However, in the newly identified study, a team of seven trained nurses performed the two procedures and obtained blood samples on the first attempt.

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ADDITIONAL TABLES

Table 1. Overview of included studies

Study ID	Country	Sample size	Sweet-tasting solution: pain relief	Outcomes
Shah 1997	UK	27	No	NIPS scores

Table 1. Overview of included studies (Continued)

				Need for > 1 skin puncture
Larsson 1998	Sweden	120	No	NFCS scores
				Need for > 1 skin puncture
Alcaraz Sanz 1998	Spain	103	No	Need for > 1 skin puncture
Eriksson 1999	Sweden	120	Yes	PIPP scores
				Need for > 1 skin puncture
Kvist 2002	Sweden	90	No	NIPS scores
				Need for > 1 skin puncture
Ogawa 2005	Japan	100	Yes	NFCS scores
				Need for > 1 skin puncture
Saththasivam 2009	Malaysia	66	Yes	NFCS scores
				Need for > 1 skin puncture
Shrestha 2012	Nepal	200	No	NIPS scores

Abbreviations: NFCS: Neonatal Facial Coding System; NIPS: Neonatal Infant Pain Scale; PIPP: Premature Infant Pain Profile; UK: United Kingdom.